

Project 1: Numerical Solutions to Elliptic PDEs

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This project implements and analyzes numerical solutions to the two-dimensional heat equation with a spatially varying source term $\kappa(x, y)$, where the steady-state solution satisfies the Poisson equation. The spatial domain is discretized using a uniform grid with second-order central finite differences, and the resulting linear system is solved using a conjugate gradient (CG) method. Mixed boundary conditions are imposed: Dirichlet conditions on the west and north boundaries, and Neumann conditions on the east and south boundaries. The transient heat equation is solved using Crank–Nicolson time integration, which is unconditionally stable and second-order accurate in time, allowing for larger timesteps than explicit methods. Marching from the initial condition $u(x, y, 0) = 0$, the transient solution converges to the steady state at $t^* = 362.4$ (3624 timesteps at $dt = 0.1$), satisfying the convergence criterion $\|u(x, y, t) - \bar{u}(x, y)\|_2 / \|\bar{u}(x, y)\|_2 < 10^{-4}$. The steady-state solution obtained from the transient simulation closely matches the direct Poisson solution, confirming the correctness of both implementations and the expected physical behavior of the system.

I. INTRODUCTION

Developed in 1822 by Joseph Fourier, the heat equation aims to model how quantities like heat diffuse through a space. It is now a fundamental concept in physics, engineering, and mathematics. In this project, we implement a numerical solution to the 2D heat equation with a spatially varying source term $\kappa(x, y)$, whose steady state solution satisfies the Poisson equation.

II. GOVERNING EQUATIONS AND BOUNDARY CONDITIONS

The heat equation of interest for this project is given by:

$$\frac{\partial u}{\partial t} = \alpha \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \kappa(x, y), \quad (1)$$

where $u(x, y, t)$ is the temperature distribution, α is the thermal diffusivity, and $\kappa(x, y)$ is the spatially varying source term. The following equation defines the source term:

$$\kappa(x, y) = 0.02 e^{-(x-0.7)^2/0.09 - (y-0.6)^2/0.25} \quad (2)$$

A heatmap of the source term $\kappa(x, y)$ is shown in Figure 1, which shows a localized heat source centered around (0.7, 0.6). The steady-state solution $\bar{u}(x, y)$ satisfies the Poisson equation:

$$\alpha \left(\frac{\partial^2 \bar{u}}{\partial x^2} + \frac{\partial^2 \bar{u}}{\partial y^2} \right) = -\kappa(x, y). \quad (3)$$

This problem uses mixed boundary conditions on the domain $[0, 1] \times [0, 1]$. The west and north boundaries have Dirichlet conditions while the east and south boundaries have Neumann conditions. The boundary conditions are

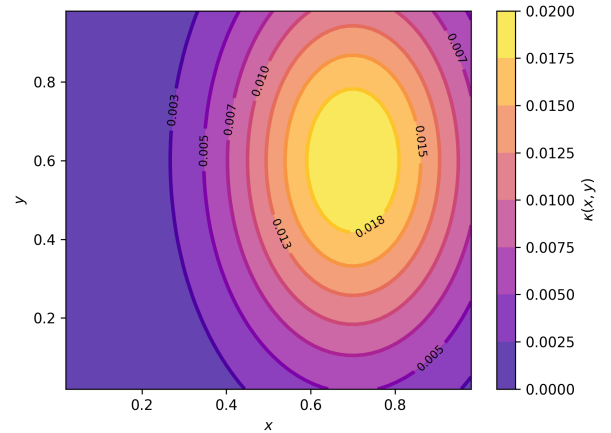


FIG. 1 Contour of source term $\kappa(x, y)$

defined in table I. Here, the west boundary condition creates a cosine variation in temperature along the y -axis, while the north boundary condition creates a sine variation along the x -axis. The Neumann conditions on the east and south boundaries specify a zero flux and a constant negative flux, respectively, which influence the overall temperature distribution and the transient behavior of the system.

TABLE I Boundary conditions applied to the square domain $[0, 1]^2$.

Boundary	Location	Type	Condition
West	$x = 0$	Dirichlet	$u = 0.5 - 0.5 \cos(2\pi y)$
East	$x = 1$	Neumann	$\partial_x u = 0$
South	$y = 0$	Neumann	$\partial_y u = -0.3$
North	$y = 1$	Dirichlet	$u = 0.5 + 0.5 \sin(4\pi x - \frac{\pi}{2})$

III. NUMERICAL METHOD

III.A. Grid and Discretization

The first step in solving this heat equation is defining the spatial grid and discretization. In this case, the domain $[0, 1] \times [0, 1]$ is discretized into a uniform grid with 50 elements in each direction. Therefore, the grid spacing is $h_x = h_y = 1/50$. This divides the domain into a total of 50×50 grid points, where the temperature u is defined at each point.

Another important aspect of this process is assigning pointers to the grid points. This gives each point a unique index that can be used to reference a specific location instead of needing to index by x and y coordinates. For example, a 3×3 grid would have the following indexing without pointers:

$$\begin{bmatrix} u(0,2) & u(1,2) & u(2,2) \\ u(0,1) & u(1,1) & u(2,1) \\ u(0,0) & u(1,0) & u(2,0) \end{bmatrix} \quad (4)$$

With pointers, the same grid would be indexed as:

$$\begin{bmatrix} u_2 & u_5 & u_8 \\ u_1 & u_4 & u_7 \\ u_0 & u_3 & u_6 \end{bmatrix} \quad (5)$$

Overall, this makes it easier to reference specific grid points when constructing the linear system for the Poisson equation and when implementing the time integration for the heat equation.

On the uniform grid, the second-order central finite difference approximation is used to discretize the spatial derivatives. The 5-point Laplacian stencil for a grid point (i, j) is given by:

$$(\mathbf{L}Q)_{i,j} = \frac{Q_{i-1,j} - 2Q_{i,j} + Q_{i+1,j}}{\Delta x^2} + \frac{Q_{i,j-1} - 2Q_{i,j} + Q_{i,j+1}}{\Delta y^2} \quad (6)$$

In the interior of the domain, this stencil is applied directly. Near the boundaries, This stencil is not directly applied since the boundary conditions must be applied. This is done by modifying the stencil to incorporate the boundary conditions, which may involve using ghost points or directly substituting the boundary values into the equations. For example, at the west boundary where $u(0, y) = 0.5 - 0.5 \cos(2\pi y)$, the stencil for the point adjacent to the boundary would be modified to include this Dirichlet condition, ensuring that the influence of the boundary is correctly represented in the numerical solution.

III.B. Linear System and Solver (Poisson)

The steady-state solution is found by solving the linear system $A\mathbf{u} = \mathbf{b}$, where A is the sparse matrix representing the discretized Laplacian operator with boundary

conditions applied, $\mathbf{u}$ is the vector of unknown temperatures at the grid points, and $\mathbf{b}$ is the vector representing the source term $\kappa(x, y)$ and contributions from boundary conditions. The conjugate gradient method is used to solve this linear system efficiently, especially given the sparsity of A . The solver is configured with a tolerance of 10^{-6} for convergence and a maximum of 1000 iterations. The stopping criterion for the solver is based on the relative residual norm, ensuring that the solution is sufficiently accurate before termination.

III.C. Time Integration (Heat Equation)

The time marching method used to solve the transient heat equation is the Crank–Nicolson method, which is an implicit time integration scheme that is second-order accurate in time and unconditionally stable. The time step dt is set to 0.1, allowing for larger time steps and fewer iterations required. The stopping criterion for the simulation is based on the L^2 norm of the difference between the transient solution $u(x, y, t)$ and the steady-state solution $\bar{u}(x, y)$ from the Poisson solver. The simulation is considered to have converged to steady state when:

$$\frac{\|u(x, y, t) - \bar{u}(x, y)\|_2}{\|\bar{u}(x, y)\|_2} < 10^{-4} \quad (7)$$

IV. RESULTS

IV.A. Poisson Steady-State Solution

Figure 2 shows the steady-state solution $\bar{u}(x, y)$ obtained by solving the Poisson equation.

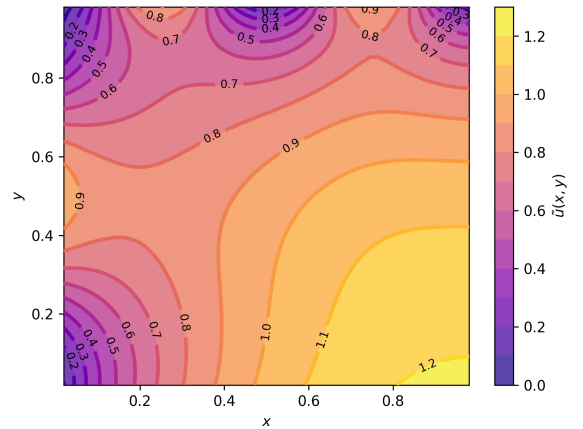


FIG. 2 Steady-state solution $\bar{u}(x, y)$ obtained by solving the Poisson equation using second-order finite differences and conjugate gradient.

Along the North boundary, the solution shows 3 sinks corresponding to the 3 peaks in the Dirichlet condition. The solution also shows a peak in the southeast corner, which is influenced by the Neumann condition and the

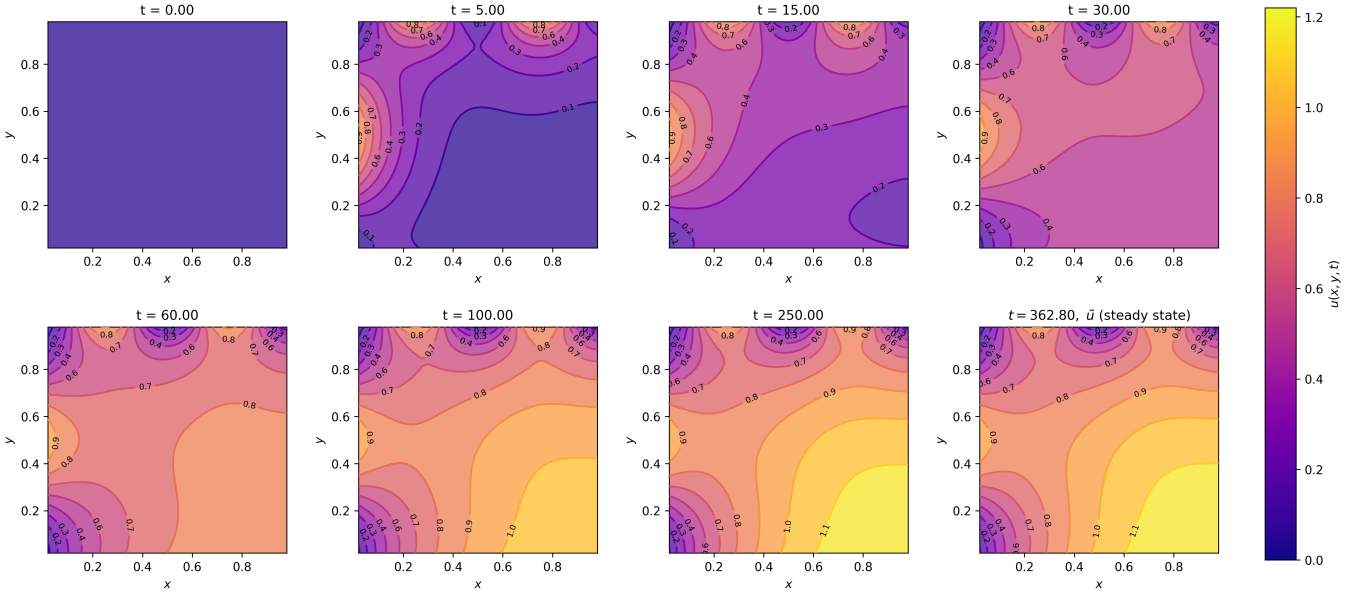


FIG. 3 Transient solution $u(x, y, t)$ at selected timesteps, marching from $u(x, y, 0) = 0$ to steady state using Crank–Nicolson time integration.

source term $\kappa(x, y)$. Furthermore, the southwest corner shows another sink, which is influenced by the Neumann condition on the south boundary and the source term.

IV.B. Transient Heat Equation and Settling Time

Figure 3 shows the transient solution $u(x, y, t)$ at selected timesteps as it marches from $u(x, y, 0) = 0$ to the steady state. The solution evolves over time, with the temperature distribution gradually changing as it approaches the steady-state solution. The transient solution converges to the steady state at $t^* = 362.4$ (3624 timesteps at $dt = 0.1$), satisfying the convergence criterion based on the L^2 norm of the difference between the transient and steady-state solutions.

IV.C. Grid Sensitivity (Optional but Recommended)

Briefly show that results are mesh-independent (e.g., 21×21 , 31×31).

V. DISCUSSION

The results confirm the expected physical behavior of a diffusion system driven by mixed boundary conditions and a localized source. The direct Poisson solution and the long-time limit of the Crank–Nicolson transient simulation are in agreement, providing validation of both implementations.

The features of the steady-state solution are governed primarily by the Dirichlet boundaries. The three sinks along the north boundary arise directly from the three full periods of $\sin(4\pi x - \pi/2)$, while the cosine variation on the west boundary establishes the large-scale gradient

that drives heat from the interior toward $x = 0$. The source $\kappa(x, y)$, centered near $(0.7, 0.6)$, produces a peak in the southeastern interior. The zero-flux condition on the east boundary ($\partial_x u = 0$) acts as a reflective wall, while the constant negative flux on the south boundary ($\partial_y u = -0.3$) draws heat out of the domain, contributing to the sink visible in the southwest corner.

The Crank–Nicolson scheme was chosen over other methods because it is unconditionally stable, allowing a large timestep $dt = 0.1$. The second-order temporal accuracy of Crank–Nicolson also ensures that the solution trajectory is well-resolved even at this larger step size. The conjugate gradient solver converged within the prescribed 10^{-6} relative residual tolerance at each timestep.

VI. CONCLUSION

This project demonstrated a complete numerical solution workflow for the two-dimensional heat equation with a spatially varying source term. The spatial domain was discretized on a 50×50 uniform grid using second-order central finite differences, yielding a sparse linear system solved by the conjugate gradient method to obtain the direct steady-state (Poisson) solution. The transient problem was integrated in time using the Crank–Nicolson scheme, which provided unconditional stability and second-order temporal accuracy, enabling a timestep of $dt = 0.1$ and convergence to steady state at $t^* = 362.4$.

The transient solution at long times closely matched the direct Poisson solution, confirming the correctness of both implementations. The solution structure—three boundary-driven sinks along the north, a localized interior peak from the Gaussian source, and a flux-driven sink in the southwest—is physically consistent with the

imposed mixed boundary conditions.

Overall, the combination of second-order spatial discretization, an efficient iterative solver, and an unconditionally stable implicit time integrator proved well-suited

to this class of elliptic and parabolic PDE problems.

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